

PROTO — Improvement Plan (Professor's 6 Points)

Sep 29, 2026 · @Ksenia

Who does what

Each point goes to the person whose part of the project it touches. Point 1 (new IRIs) must be done **first**, because every other file will use the new IRIs.

#	Professor's request	Owner	Files touched	Starts when
1	Ontology and element IRIs from the GitHub raw file	Sevda (with Saba for the TBox)	TBox, ABox, <code>xlsx_to_rdf.py</code> , TriplyDB	Day 1
4	<code>owl:imports</code> of the schema into the knowledge graph	Sevda	ABox header, TriplyDB	Right after 1
3	Missing property types (ObjectProperty / DatatypeProperty)	Saba	TBox (+ ABox declarations)	After 1
5	Class restrictions and disjoint classes	Saba	TBox	After 3
2	Complete annotations for the ontology and its elements	Claudia	TBox header + every element, WIDOCO, website	After 1; final pass after 5
6	Test the ontology by extracting proverbial knowledge/bias from texts	Polyxeni	New file <code>test/</code> + new SPARQL queries	Now (text collection), graph after 4

Why the professor saw missing types (point 3). In the TBox, `linksProverbA`, `linksProverbB` and `hasEquivalenceType` are declared as `owl:ObjectProperty`. But `proto_data.ttl` (the knowledge graph) has no ontology header and does not import the TBox. Opened alone, e.g. in Protégé, those properties have no type. So points 3 and 4 fix the same problem from two sides.

Sevda — IRIs (point 1) and owl:imports (point 4)

Today everything uses `http://example.org/proto#`, a placeholder that does not resolve. The professor wants IRIs that point to the real files on GitHub.

Suggested IRIs (confirm the branch name is `main`):

What	New IRI
Schema (TBox) ontology	<code>https://raw.githubusercontent.com/callmesevda/KR-Project/main/proto-ontology-structure.ttl</code>
Schema elements (prefix <code>proto:</code>)	<code>https://raw.githubusercontent.com/callmesevda/KR-Project/main/proto-ontology-structure.ttl#</code>
Knowledge graph (ABox) ontology	<code>https://raw.githubusercontent.com/callmesevda/KR-Project/main/proto_data.ttl</code>
Individuals (proverbs, links)	<code>https://raw.githubusercontent.com/callmesevda/KR-Project/main/proto_data.ttl#</code>

With this split, classes and properties live in the schema file and individuals (`PRV-001`, `Equiv_PRV-001_PRV-002` ...) live in the data file.

Tasks

- ☐ Change the `@prefix` and the ontology IRI in `proto-ontology-structure.ttl` (coordinate with Saba, who edits the same file).
- ☐ In `xlsx_to_rdf.py`, replace `PROTO = Namespace("http://example.org/proto#")` with the schema IRI, and add a second namespace for individuals.
- ☐ Add an ontology header with `owl:imports` to the ABox (can be generated by the script, see below).
- ☐ Regenerate `proto_data.ttl`, copy both files into `docs/data/`, re-upload to TriplyDB.
- ☐ Tell Polyxeni and Claudia when done: queries and website must switch prefix.

Header to add to the ABox:

```
@prefix proto: <https://raw.githubusercontent.com/callmesevda/KR-
Project/main/proto-ontology-structure.ttl#> .
@prefix : <https://raw.githubusercontent.com/callmesevda/KR-
Project/main/proto_data.ttl#> .

<https://raw.githubusercontent.com/callmesevda/KR-
Project/main/proto_data.ttl>
  a owl:Ontology ;
  owl:imports <https://raw.githubusercontent.com/callmesevda/KR-
Project/main/proto-ontology-structure.ttl> ;
  rdfs:label "PROTO knowledge graph" .
```

Test: open the raw URL of `proto_data.ttl` in Protégé (*File* → *Open from URL*). The schema should load automatically, and `linksProverbA` should appear under Object Properties.

Saba — property types (point 3), restrictions and disjoint classes (point 5)

Point 3 — types. Every property used anywhere must be declared in the TBox with its type:

Property	Type	Status now
<code>hasLesson</code> , <code>hasSourceDomain</code> , <code>usedIn</code> , <code>linksProverbA</code> , <code>linksProverbB</code> , <code>hasEquivalenceType</code> , <code>groundedIn</code>	<code>owl:ObjectProperty</code>	Declared in TBox; missing when ABox is read alone (fixed by point 4)
<code>hasText</code> , <code>hasTranslation</code> , <code>hasProverbID</code> , <code>hasLanguage</code> , <code>hasLabel</code> , <code>hasDivergenceNote</code> , <code>annotatedBy</code>	<code>owl:DatatypeProperty</code>	Declared in TBox
<code>groundedIn</code>	ObjectProperty, but has no range	Add a range, or document that it points to external Wikidata entities

Also check, in Protégé with the ABox loaded, that no property appears under *Annotation properties* by mistake.

Point 5 — restrictions and disjointness. Goal: when bad data enters the graph, the reasoner (HermiT in Protégé) must report it. Suggested axioms:

1. **Disjoint classes:** `Proverb`, `TacitLesson`, `SituationOfUse`, `SourceDomain`, `CrossCulturalEquivalence`, `EquivalenceType` are all pairwise disjoint (`owl:AllDisjointClasses`).
2. **Closed list of equivalence types:** `EquivalenceType` $\equiv$ `{DirectMatch, SameLessonDifferentImage, PartialOverlap, FalseFriend}` (`owl:oneOf`) + `owl:AllDifferent` on the four.
3. **Functional properties:** `hasProverbID`, `hasLanguage`, `hasText`, `linksProverbA`, `linksProverbB`, `hasEquivalenceType`.
4. **Class restrictions on `Proverb`:** `hasLesson` exactly 1 `TacitLesson`, `hasSourceDomain` some `SourceDomain`, `usedIn` some `SituationOfUse`.
5. **Class restrictions on `CrossCulturalEquivalence`:** `linksProverbA` exactly 1 `Proverb`, `linksProverbB` exactly 1 `Proverb`, `hasEquivalenceType` exactly 1 `EquivalenceType`.

Test cases to show the professor. Add each wrong triple in a copy of the file, run the reasoner, and screenshot the inconsistency:

- a proverb typed also as a `TacitLesson` → disjointness violation;
- a link with two different equivalence types → functional property + `AllDifferent` violation;
- a link whose `linksProverbA` points to a lesson → range + disjointness violation.

Note: OWL is open-world, so a *missing* value (a proverb without a lesson) will **not** be flagged. Say this in the documentation; Polyxeni's SPARQL checks cover it.

Claudia — annotations (point 2), WIDOCO and website

Today most metadata (authors, license, dates, abstract) lives only in `widoco-config.properties`. The TBox itself has 10 `rdfs:comment`s and no author, license or version. The professor wants it **inside the ontology**.

Ontology header — add to the TBox (and a shorter one to the ABox header Sevda creates):

```
<https://raw.githubusercontent.com/callmesevda/KR-Project/main/proto-ontology-structure.ttl>
  a owl:Ontology ;
  dcterms:title "PROTO — Proverbs & Tacit-knowledge Ontology"@en ;
  dcterms:description "..."@en ;
  dcterms:creator "Sevda Rezaei Melal", "Polyxeni Chasanai",
                  "Saba Afsharzadehtorghab", "Claudia Briccolani" ;
  dcterms:created "2026-09-10"^^xsd:date ;
  dcterms:modified "...^^xsd:date ;
  dcterms:license <https://creativecommons.org/licenses/by/4.0/> ;
  owl:versionInfo "1.1.0" ;
  owl:versionIRI <...> ;
  vann:preferredNamespacePrefix "proto" ;
  vann:preferredNamespaceUri
"https://raw.githubusercontent.com/callmesevda/KR-Project/main/proto-ontology-structure.ttl#" .
```

Every element (classes, properties, the 4 equivalence types) should have:

- ☐ `rdfs:label` with a language tag (`"Proverb"@en`);
- ☐ `rdfs:comment` — a one-sentence definition. Missing today on 11 of the 14 properties (only ``groundedIn``, ``hasTranslation``, ``hasProverbID`` have one) and on `SameLessonDifferentImage`, `FalseFriend`, `PartialOverlap`, `EquivalenceType` ;
- ☐ optionally `skos:example` (e.g. a proverb ID) and `rdfs:isDefinedBy` pointing to the ontology IRI.

Definition for PartialOverlap (it has none): take it from Polyxeni's review — *"same lesson; the images share an element or scene type without being the same image."*

Then: regenerate WIDOCO from the new TBox (after Saba's point 5 is in), and update the website: new prefix in the example queries, new IRIs on the Database page, and a short "What changed in v1.1" section.

Polyxeni — testing the ontology on real texts (point 6) + queries

The idea. Take real texts where people *use* proverbs, and turn them into a small knowledge graph using PROTO as the lens: which proverb, what lesson, in what situation, applied to whom. If the ontology can describe these uses, it passes the test. What it *cannot* describe is also a result: it shows where to extend it. This is the "Extraction" half of the course name.

Steps

1. **Collect 6–8 short texts** containing a proverb, spread across genres and languages: 2 news articles, 2 blog or opinion posts, 2 chat or forum excerpts (e.g. Reddit, comments). Priority: proverbs already in the corpus ("one bad apple", "still waters", "chi cerca trova", "what goes around..."). Save source URL, date, and the quoted passage.
2. **Annotate by hand (gold standard)**. For each occurrence: the proverb (existing `PRV-xxx`, or a new one), its tacit lesson, and a new `SituationOfUse` individual whose label describes *this* use, with `dcterms:source` = the URL.
3. **Look for bias**. Note what the proverb is applied to and how. Example: "one bad apple" used about a whole group (police, migrants, a company) as a generalisation; "still waters" used positively about one person, negatively about another. Record this in the situation label or in a note.
4. **Automatic extraction (optional but strong)**. Give an LLM the TBox plus each text and ask it to output Turtle using only PROTO's classes and properties. Compare with the hand annotation: how many occurrences it found, how many lessons it got right.
5. **Validate** the extracted graph: it imports the TBox, so Saba's restrictions apply; run the reasoner and the SPARQL checks below.
6. **Write up** half a page: what the ontology captured, what it missed (e.g. no property for *who* the proverb targets, no way to mark irony or negative stance), and the proposed extension (e.g. `ProverbOccurrence`, `appliedTo`, `hasStance`).

Queries to update or add

- ☐ Switch the 8 competency queries to the new prefix after Sevda's point 1, and re-run them on TriplyDB.
- ☐ Add SPARQL checks for what OWL cannot catch (open world): a proverb with no lesson; an equivalence joining two proverbs of the same language; a link whose two proverbs have different lessons.
- ☐ Add 1–2 queries over the text graph, e.g. "which proverbs are used in news vs chats" or "which lessons are applied to groups of people".

Deliverables: `test/texts.md` (sources + quotes), `test/text_kg.ttl` (imports the TBox), `test/queries.rq`, and a short "Evaluation" section for Claudia to add to the website.

Open questions for the group

- **Deadline?** The professor gave none. We should set one and tell him when v1.1 is ready.
- **IRI scheme:** one namespace for everything, or schema `#` and data `#` separately (as proposed above)? Sevda and Saba decide before anyone else starts.

- **Apply Polyxeni's reclassification now?** The IRI change requires regenerating the ABox anyway. It could be the moment to also set the reviewed types and `annotatedBy` on the 48 links (currently future work).
- **Extend the ontology after point 6?** If the text test shows gaps (targets, stance), do we add them in v1.1 or only list them as future work?
- **Language codes:** fix `ita / alb` → `it / sq` while regenerating the data?